TCSS 321 – Discrete Structures

Propositional Logic

Propositions

Definition: A proposition is a figure of speech that can be either true or false.

- "Snow is white."
- "1 + 1 = 2."
- "The President of the United States is Kanye West."

- Other figures of speech do not have truth values.
  - "Go away!"
  - "Where is the bathroom?"
  - "Wubalubadubdub!"

Definition: A tautology is a proposition that is always true.

- "I am over 21 or I am not over 21."

Definition: A contradiction is a proposition that is never true.

- "I am over 21 and I am not over 21."

- In propositional logic we represent propositions with lowercase letters like $p, q, r, s, \ldots$ (sometimes $a, b, c, \ldots$).

Compound Propositions

- Propositions can be combined to create new propositions using logical connectives.
  - "Snow is white and the President of United States is not Kanye West."
  - "If it is sunny outside, then we will go for a picnic."
  - "I am neither a pirate nor a ninja."

Name	Formal Name	Logical Symbol	Java Symbol	Other Symbols
Not	Negation	$\neg p$	$!\,p$	$\overline{p}$
And	Conjunction	$p \wedge q$	$p \,\&\&\, q$	$p \cdot q, p \times q$
(Inclusive) Or	Inclusive Disjunction	$p \vee q$	$p \,\|\|\, q$	$p + q$
(Exclusive) Or	Exclusive Disjunction	$p \oplus q$	None	None
If … Then …	Implication	$p \rightarrow q$	None	None
… only if …	Converse Implication	$p \leftarrow q$	None	None
If and only if	Biconditional	$p \leftrightarrow q$	None	None

Truth Tables

- We can define exactly how the logical connectives operate by stating the truth table for each connective.

p	$\neg p$
T	F
F	T

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

p	q	$p \oplus q$
T	T	F
T	F	T
F	T	T
F	F	F

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

Inclusive and Exclusive Or (Disjunction)

- The use of "or" in normal speech is ambiguous.

- "You must be at least 12 years old or accompanied by an adult."
  - This statement uses inclusive or which means "Either p or q or both.".

- "Your meal comes with a soup or a salad."
  - This statement uses exclusive or which means "Either p or q but not both.".

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

p	q	$p \oplus q$
T	T	F
T	F	T
F	T	T
F	F	F

(Converse) Implication and Biconditional

- The use of "If … then …" can also be ambiguous in normal speech.

- "If you get a 4.0 then you will graduate."
  - This is a standard implication which means "if p then q but if not p then either q or not q.".

- "If you take a shot then you can score a goal."[1] – Wayne Gretzky (paraphrase)
  - This is a converse implication which means "if p then either q or not q but if not p then not q.".

- "If you finish your vegetables then you can have ice cream."
  - This is a biconditional which means "if p then q but if not p then not q.".

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

p	q	$p \leftarrow q$
T	T	T
T	F	T
F	T	F
F	F	T

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

Logical Equivalence

- We can show that exclusive or can be rewritten using only inclusive or, negation and conjunction by truth table evaluation.
  - We will show that $p \oplus q \equiv (p \vee q) \wedge \neg(p \wedge q)$ where $\equiv$ means "is logically equivalent", or "has the same truth table".

$\boldsymbol{p}$	$\boldsymbol{q}$	$\boldsymbol{p \oplus q}$	$\boldsymbol{p \wedge q}$	$\boldsymbol{p \vee q}$	$\boldsymbol{\neg(p \wedge q)}$	$\boldsymbol{(p \vee q) \wedge \neg(p \wedge q)}$
T	T	F	T	T	F	F
T	F	T	F	T	T	T
F	T	T	F	T	T	T
F	F	F	F	F	T	F

Logical Equivalence

- Can you show that $p \oplus q \equiv \neg(p \leftrightarrow q)$ using truth table evaluation?

$\boldsymbol{p}$	$\boldsymbol{q}$	$\boldsymbol{p \oplus q}$	$\boldsymbol{p \leftrightarrow q}$	$\boldsymbol{\neg(p \leftrightarrow q)}$
T	T	F	T	F
T	F	T	F	T
F	T	T	F	T
F	F	F	T	F

Logical Equivalence

- Can you show that $p \leftrightarrow q \equiv (p \rightarrow q) \wedge (p \leftarrow q)$ using truth table evaluation?

$\boldsymbol{p}$	$\boldsymbol{q}$	$\boldsymbol{p \leftrightarrow q}$	$\boldsymbol{p \rightarrow q}$	$\boldsymbol{p \leftarrow q}$	$\boldsymbol{(p \rightarrow q) \wedge (p \leftarrow q)}$
T	T	T	T	T	T
T	F	F	F	T	F
F	T	F	T	F	F
F	F	T	T	T	T

de Morgan's Laws

- There are two useful equivalences called de Morgan's Laws.

$$\neg(p \wedge q) \equiv (\neg p \vee \neg q)$$
$$\neg(p \vee q) \equiv (\neg p \wedge \neg q)$$

Can we prove these logical equivalences using truth tables?

p	q	$\neg p$	$\neg q$	$\neg(p \wedge q)$	$(\neg p \vee \neg q)$	$\neg(p \vee q)$	$(\neg p \wedge \neg q)$
T	T	F	F	F	F	F	F
T	F	F	T	T	T	F	F
F	T	T	F	T	T	F	F
F	F	T	T	T	T	T	T

Logical Equivalences

- There are many logical equivalences that we can prove with truth tables.

Logical Equivalence	Arithmetic Equivalence	Name
$\neg(\neg p) \equiv p$	$-(-a) = a$	Double negation
$p \wedge T \equiv p$ $p \vee F \equiv p$	$a \cdot 1 = a$ $a + 0 = a$	Identity
$p \vee T \equiv T$ $p \wedge F \equiv F$	$a \cdot 0 = 0$	Domination
$p \wedge q \equiv q \wedge p$ $p \vee q \equiv q \vee p$	$a \cdot b = b \cdot a$ $a + b = b + a$	Commutative
$(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$ $(p \vee q) \vee r \equiv p \vee (q \vee r)$	$(a \cdot b) \cdot c = a \cdot (b \cdot c)$ $(a + b) + c = a + (b + c)$	Associative
$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$ $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$	$a \cdot (b + c) = (a \cdot b) + (a \cdot c)$	Distributive

Logical Equivalences

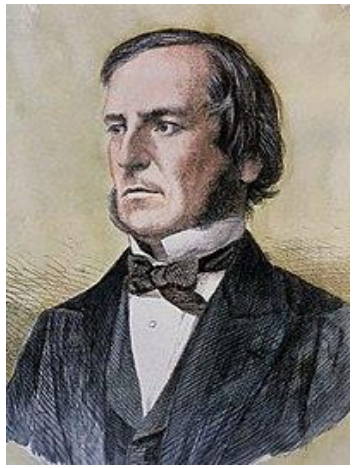
- There are many logical equivalences that we can prove with truth tables.

Logical Equivalence	Name
$p \wedge p \equiv p$ $p \vee p \equiv p$	Idempotent
$\neg(p \wedge q) \equiv \neg p \vee \neg q$ $\neg(p \vee q) \equiv \neg p \wedge \neg q$	De Morgan's
$p \vee (p \wedge q) \equiv p$ $p \wedge (p \vee q) \equiv p$	Absorption
$p \vee \neg p \equiv T$ $p \wedge \neg p \equiv F$	Negation
$p \rightarrow q \equiv \neg q \rightarrow \neg p$	Contrapositive

Boolean Algebra

- If a logical formula has n different propositions how many rows will the truth table have?

$$2^n$$



- Say we want to show that $\neg\big((p \vee q) \vee (r \vee s)\big) \equiv \neg p \wedge \neg q \wedge \neg r \wedge \neg s$. Can we do this without using a truth table? Hint: Use De Morgan's law.

$$\begin{aligned}\neg\big((p \vee q) \vee (r \vee s)\big) &\equiv \neg(p \vee q) \wedge \neg(r \vee s)\\ &\equiv \neg p \wedge \neg q \wedge \neg(r \vee s)\\ &\equiv \neg p \wedge \neg q \wedge \neg r \wedge \neg s\end{aligned}$$

- These logical formulas are called Boolean formulas and this type of algebra is called Boolean algebra.
  - They were named after English mathematician George Boole (1815-1864) as was the truth having type Boolean (or bool).

Boolean Algebra

- Can we show that $\neg\big(p \vee (\neg p \wedge q)\big) \equiv \neg p \wedge \neg q$ using Boolean algebra?

$$\begin{aligned}
\neg\big(p \vee (\neg p \wedge q)\big) &\equiv \neg p \wedge \neg(\neg p \wedge q) \\
&\equiv \neg p \wedge (\neg(\neg p) \vee \neg q) \\
&\equiv \neg p \wedge (p \vee \neg q) \\
&\equiv (\neg p \wedge p) \vee (\neg p \wedge \neg q) \\
&\equiv F \vee (\neg p \wedge \neg q) \\
&\equiv \neg p \wedge \neg q
\end{aligned}$$

TCSS 321 – Discrete Structures

Sets

Sets

Definition: A set is a collection of elements (or members).

- The set of all Presidents of the United States.

$$\{\text{George Washington}, \dots, \text{Barack Obama}, \text{Donald Trump}\}$$

- The set of natural numbers.

$$\mathbb{N} = \{0,1,2, \dots\}$$

- The set of characters in the English alphabet.

$$\{a, b, c, d, e, f, g, h, i, j, k, l, m, n, o, p, q, r, s, t, u, v, w, x, y, z\}$$

- The set of Harry Potter books.

$$\{\text{Philosopher's Stone}, \dots, \text{The Deathly Hallows}\}$$

- We use curly braces {} to enclose sets.
- Sets are unordered collections (though we sometimes state them in a conventional order).
  - $\{1,2,3\} = \{3,2,1\} = \{2,1,3\}$
- Sets do not permit duplicates.
  - If we allow a set to have duplicates, we call it a multi-set.

Set Membership

- All that matters to a set is which elements are counted as members.

- We use the notations $s \in S$ to mean "s is an element of S".

Example: Let $\mathbb{E} = \{0,2,4, \dots\}$ be the set of even natural numbers.

- Then $4 \in \mathbb{E}$ and $5 \notin \mathbb{E}$.

Example: Let $V = \{a, e, i, o, u\}$ be the set of vowels.

- Then $a \in V$ and $b \notin V$.

Sets of Numbers

- There are several sets of numbers we are already familiar with.

$\mathbb{N} = \{0,1,2,3, \dots\}$, the set of natural numbers
$\mathbb{Z} = \{\dots, -2, -1,0,1,2, \dots\}$, the set of integers
$\mathbb{Z}^+ = \{1,2, \dots\}$, the set of positive integers
$\mathbb{Q} = \left\{\frac{p}{q} | p \in \mathbb{Z}, q \in \mathbb{Z}, \text{and } q \neq 0\right\}$, the set of rational numbers
$\mathbb{R} = (-\infty, \dots, \infty)$, the set of real numbers
$\mathbb{C} = \{\alpha + \beta i \mid \alpha \in \mathbb{R} \wedge \beta \in \mathbb{R}\}$, the set of complex numbers

- To define the set of rational numbers (and complex numbers) we used set builder notation.

$$\{x | \ x \text{ has some property}\}$$

read as "the set of all x such that x has some property".

Subsets and Supersets

Definition: A set A is a subset of a set B, denoted $A \subseteq B$, if $a \in A$ then $a \in B$. We also say that B is a superset of A, denoted $B \supseteq A$, if $A \subseteq B$.

- We can see that some sets of numbers are a subset of others.

$$\mathbb{Z}^+ \subseteq \mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R} \subseteq \mathbb{C}$$

- The notation $\subseteq$ means "subset or equal". The notation $\subset$ means "proper subset" or "subset but not equal".

$$\mathbb{Z}^+ \subset \mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$$

Example: Let $A = \{a, b, c, \dots, x, y, z\}$ be the set of English alphabet symbols and let $V = \{a, e, i, o, u\}$ be the set of vowels.

$$V \subset A$$

The Universal Set and The Empty Set

- The universal set defaults to the set of all "things".
  - We leave "things" here deliberately vague.
  - The universal set includes all numbers, all characters, all particles in the physical universe, all people (past, present and future), all dragons,…
  - Sometimes for convenience we will restrict the universal set to a subset like all numbers.

Definition: The universal set U is the set such that every set $S \neq U$ is such that $S \subseteq U$.

Definition: The empty set, denoted as $\emptyset$ (also sometimes $\{\}$), is the set with no members.

Theorem: If S is a set then $\emptyset \subseteq S$.

Proof: To show that $\emptyset \subseteq S$ we must show that if $x \in \emptyset$ then $x \in S$. Since there are no $x \in \emptyset$ the antecedent is false and thus the conditional is true. ■

The Size of a Set

Definition: The size of a set S, denoted $|S|$, is equal to the number of members of the set. The size of a set is also known as the cardinality of a set.

- The size of a set is always a natural number or infinity.
  - If the size of a set is a natural number, we call the set finite.
  - Otherwise the set is infinite.

Examples:

- $|\emptyset| = 0$
- $|\mathbb{N}| = \infty$
- $|\{x \in \mathbb{N} | x \leq 10\}| = 11$

Power Sets

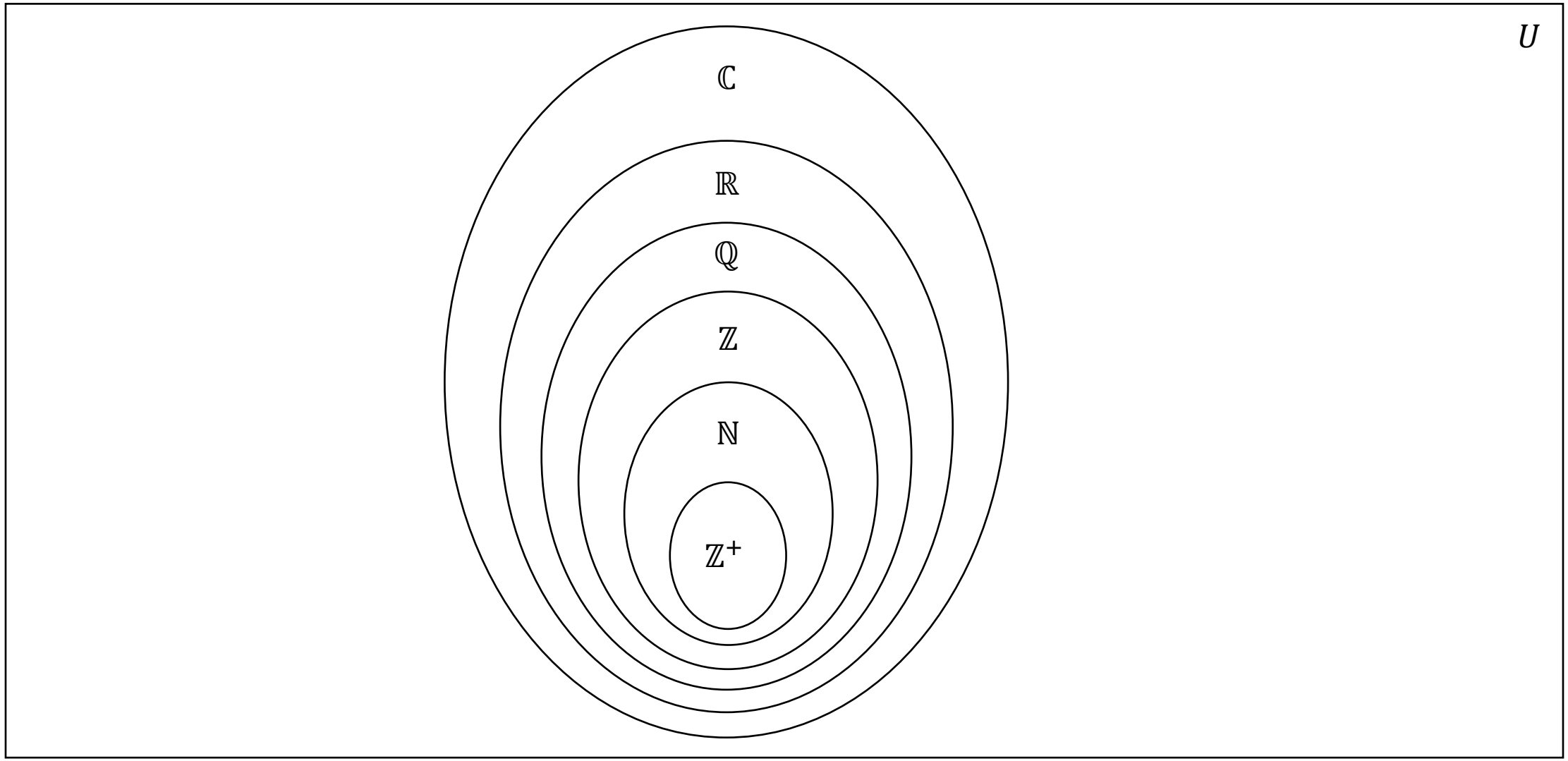
Definition: Let S be a set. The power set of S, denoted $P(S)$, is the set of all subsets of S.

- Let $S = \{1,2,3,4\}$. List all the subsets of S.

$\{\}$	$\{1\}$	$\{2\}$	$\{3\}$
$\{4\}$	$\{1,2\}$	$\{1,3\}$	$\{1,4\}$
$\{2,3\}$	$\{2,4\}$	$\{3,4\}$	$\{1,2,3\}$
$\{1,2,4\}$	$\{1,3,4\}$	$\{2,3,4\}$	$\{1,2,3,4\}$

- If $|S| = n$, then what is $|P(S)|$?
  - For each of the n elements we have 2 options: either the element is in the subset or it is not.
  - This means we have $\overbrace{2 \cdot 2 \cdots 2}^{n} = 2^n$ subsets
  - This is called the product rule in counting (see section 6.1 of the text).

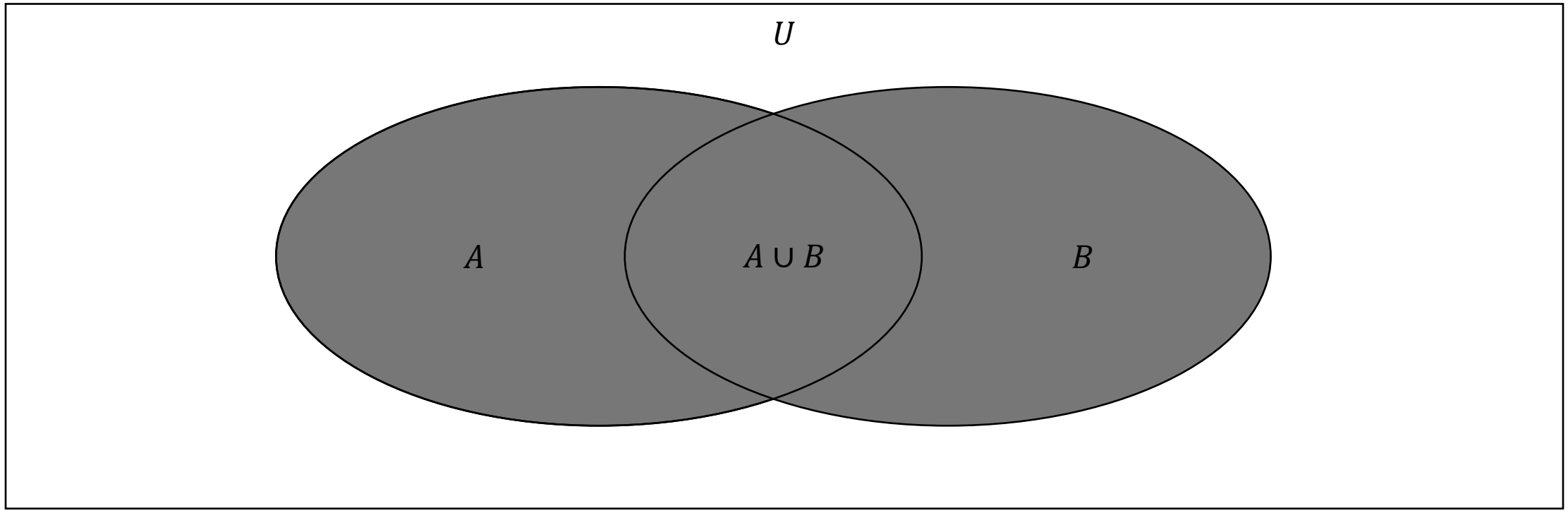
Venn Diagram



Union

Definition: Let A and B be sets. Then the union of A and B is

$$A \cup B = \{x \mid x \in A \vee x \in B\}$$



Union

- Let $\mathbb{E} = \{0,2,4, \dots\} = \{2k \mid k \in \mathbb{N}\}$ be the even natural numbers and let $\mathbb{O} = \{1,3,5, \dots\} = \{2k + 1 \mid k \in \mathbb{N}\}$ be the odd natural numbers. What is $\mathbb{E} \cup \mathbb{O}$?

$$\mathbb{E} \cup \mathbb{O} = \mathbb{N}$$

- Can we prove this? To show that sets $A = B$ we need to show that $A \subseteq B$ and $A \supseteq B$.

Lemma 1: $\mathbb{E} \cup \mathbb{O} \subseteq \mathbb{N}$

Proof: To prove that $\mathbb{E} \cup \mathbb{O} \subseteq \mathbb{N}$ we need to show that if $x \in \mathbb{E} \cup \mathbb{O}$ then $x \in \mathbb{N}$. If $x \in \mathbb{E} \cup \mathbb{O}$ then $x \in \mathbb{E}$ or $x \in \mathbb{O}$ by definition.

Case 1: $x \in \mathbb{E}$. If $x \in \mathbb{E}$ then x is an even natural number. Thus x is a natural number and $x \in \mathbb{N}$.

Case 2: $x \in \mathbb{O}$. If $x \in \mathbb{O}$ then x is an odd natural number. Thus x is a natural number and $x \in \mathbb{N}$.

We have proven both cases so if $x \in \mathbb{E} \cup \mathbb{O}$ then $x \in \mathbb{N}$ and therefore $\mathbb{E} \cup \mathbb{O} \in \mathbb{N}$. ■

- This is an example of a direct proof and an exhaustive proof.

Union

Lemma 2: $\mathbb{N} \subseteq \mathbb{E} \cup \mathbb{O}$

Proof: To prove that$\mathbb{N} \subseteq \mathbb{E} \cup \mathbb{O}$ we need to show that if $x \in \mathbb{N}$ then $x \in \mathbb{E} \cup \mathbb{O}$. If $x \in \mathbb{N}$ then $x = 2k$ or $x = 2k + 1$.

Case 1: $x = 2k$. If $x = 2k$ then x is an even natural number. Thus $x \in \mathbb{E}$.

Case 2: $x = 2k + 1$. If $x = 2k + 1$ then x is an odd natural number. Thus $x \in \mathbb{O}$.

We have proven both cases so if $x \in \mathbb{E} \cup \mathbb{O}$ then $x \in \mathbb{N}$ and therefore $\mathbb{E} \cup \mathbb{O} \in \mathbb{N}$. ■

- A lemma is an intermediary proposition we prove in order to prove a different theorem.

Theorem: $\mathbb{N} = \mathbb{E} \cup \mathbb{O}$.

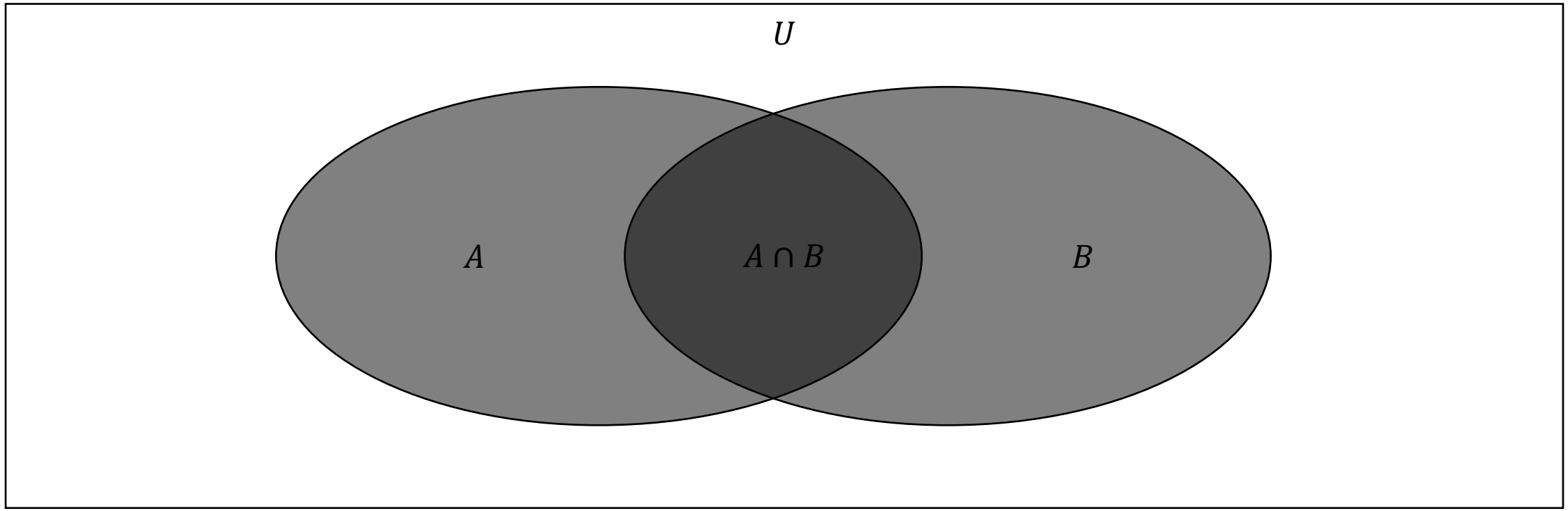
Proof: By definition $\mathbb{N} = \mathbb{E} \cup \mathbb{O}$ if $\mathbb{N} \subseteq \mathbb{E} \cup \mathbb{O}$ and $\mathbb{N} \supseteq \mathbb{E} \cup \mathbb{O}$. By Lemma 1 and Lemma 2 the theorem follows. ■

- When a theorem is an immediate consequence of other propositions it is called a corollary.

Intersection

Definition: Let A and B be sets. Then the intersection of A and B is

$$A \cap B = \{x \mid x \in A \wedge x \in B\}$$



Intersection

- Let $\mathbb{E} = \{0,2,4, \dots\} = \{2k \mid k \in \mathbb{N}\}$ be the even natural numbers and let $\mathbb{O} = \{1,3,5, \dots\} = \{2k + 1 \mid k \in \mathbb{N}\}$ be the odd natural numbers. What is $\mathbb{E} \cap \mathbb{O}$?

$$\mathbb{E} \cap \mathbb{O} = \emptyset$$

- If $A \cap B = \emptyset$ we say that A and B are disjoint.

- $\mathbb{E}$ and $\mathbb{O}$ are an example of a partitioning of $\mathbb{N}$.

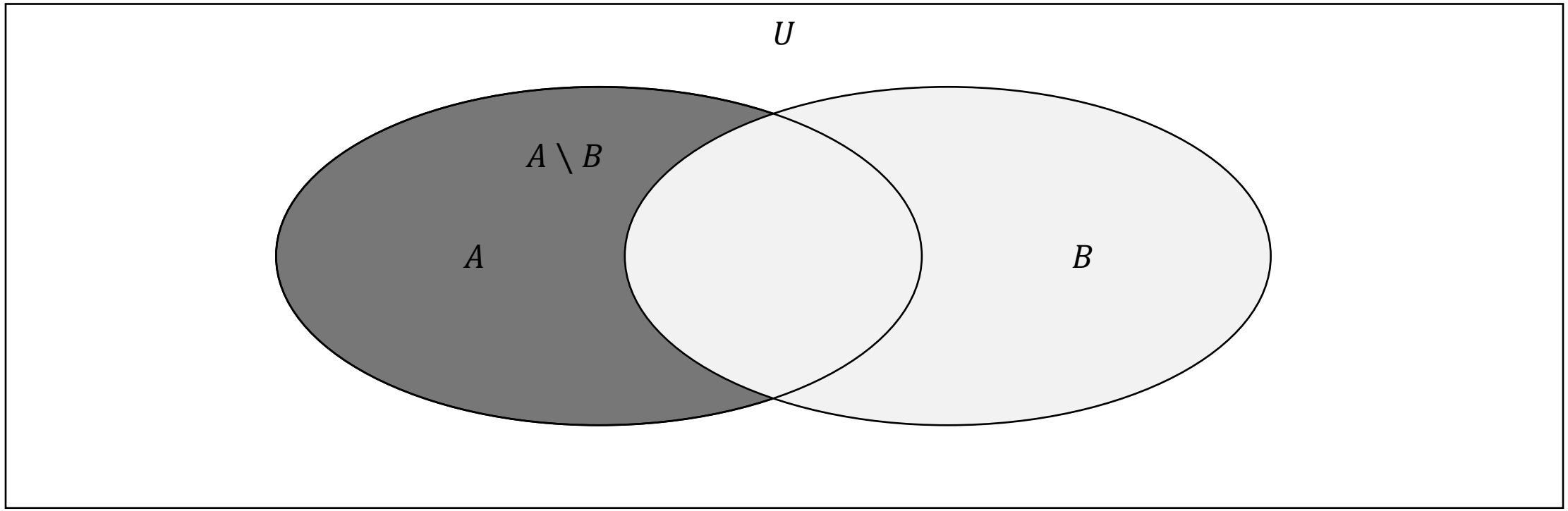
Definition: A and B partition S if $A \subseteq S$ and $B \subseteq S$ and $A \cap B = \emptyset$ and $A \cup B = S$.

- This definition can be extended to multiple subsets as long as they are all pairwise disjoint and together exhaustive.

Set Difference

Definition: Let A and B be sets. Then the set difference of A and B is

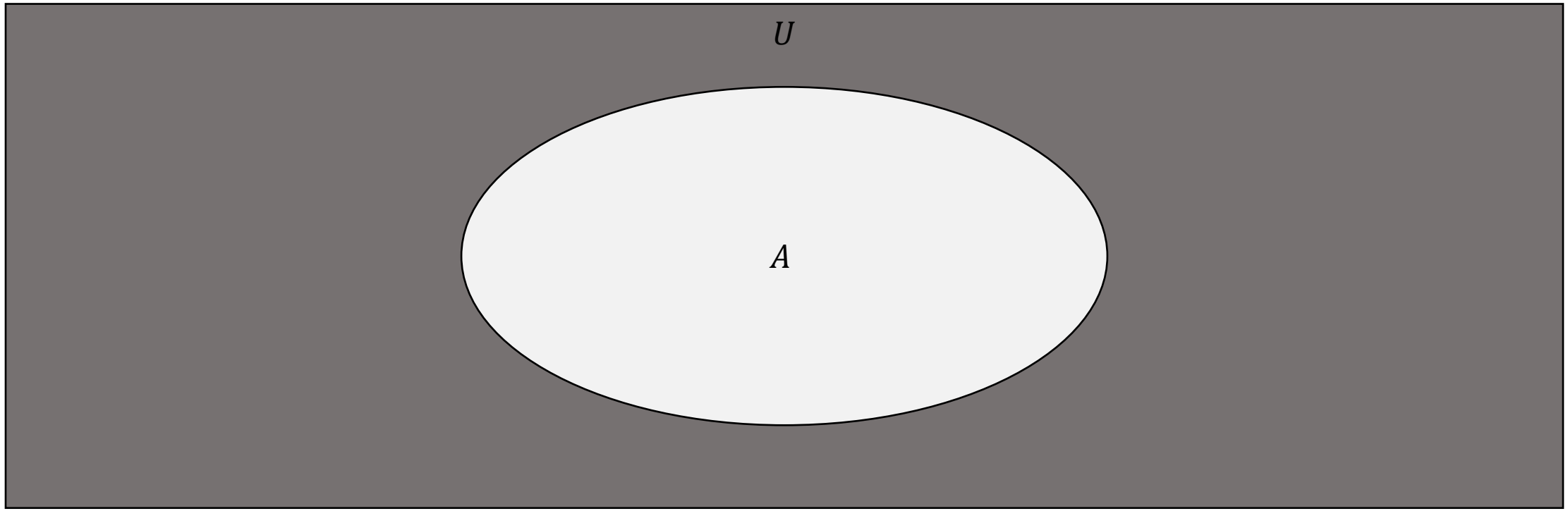
$$A \setminus B = \{x \mid x \in A \wedge x \notin B\}$$



Complement

Definition: Let A be a set. Then the complement of A is

$$\overline{A} = \{x \mid x \notin A\}$$



Set Equivalences

- There are set equivalences that parallel the logical and arithmetic equivalences.

Set Equivalence	Logical Equivalence	Arithmetic Equivalence	Name
$\overline{(\bar{A})} = A$	$\neg(\neg p) \equiv p$	$-(-a) = a$	Double negation
$A \cap U = A$ $A \cup \emptyset = A$	$p \wedge T \equiv p$ $p \vee F \equiv p$	$a \cdot 1 = a$ $a + 0 = a$	Identity
$A \cup U = U$ $A \cap \emptyset = \emptyset$	$p \vee T \equiv T$ $p \wedge F \equiv F$	$a \cdot 0 = 0$	Domination
$A \cap B = B \cap A$ $A \cup B = B \cup A$	$p \wedge q \equiv q \wedge p$ $p \vee q \equiv q \vee p$	$a \cdot b = b \cdot a$ $a + b = b + a$	Commutative
$A \cap (B \cap C) = (A \cap B) \cap C$ $A \cup (B \cup C) = (A \cup B) \cup C$	$(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$ $(p \vee q) \vee r \equiv p \vee (q \vee r)$	$(a \cdot b) \cdot c = a \cdot (b \cdot c)$ $(a + b) + c = a + (b + c)$	Associative
$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$	$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$ $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$	$a \cdot (b + c) = (a \cdot b) + (a \cdot c)$	Distributive

Set Equivalences

- There are set equivalences that parallel the logical and arithmetic equivalences.

Set Equivalence	Logical Equivalence	Name
$A \cap A = A$ $A \cup A = A$	$p \wedge p \equiv p$ $p \vee p \equiv p$	Idempotent
$\overline{(A \cap B)} = \bar{A} \cup \bar{B}$ $\overline{(A \cup B)} = \bar{A} \cap \bar{B}$	$\neg(p \wedge q) \equiv \neg p \vee \neg q$ $\neg(p \vee q) \equiv \neg p \wedge \neg q$	De Morgan's
$A \cup (A \cap B) = A$ $A \cap (A \cup B) = A$	$p \vee (p \wedge q) \equiv p$ $p \wedge (p \vee q) \equiv p$	Absorption
$A \cup \bar{A} = U$ $A \cap \bar{A} = \emptyset$	$p \vee \neg p \equiv T$ $p \wedge \neg p \equiv F$	Negation